



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

APPLIED CRYPTOGRAPHY [IT-3251]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1)
 - a. Compute all residues of Modulo 8 and its multiple of 3.
 - b. Illustrate extended Euclidian algorithm for finding the inverse of $1264 \bmod 79236$. (5)
 - A)
 - B)
 - a. Compute $J(610/987)$ $J(20964/1987)$ $J(1234567/111111)$
 - b. Investigate the relationship between Jacobi symbols and the Solvay-Strassen primality test with an illustration. (5)
- 2) Dexter wants to set up his own public and private keys. He chooses $p = 23$ and $q = 19$ with $e = 283$. Perform encryption and decryption using RSA for a message 27. (5)
 - A)
 - B) Consider the Paillier cryptosystem with the prime pair $p=17$ and $q=19$, and a chosen plaintext message $m=42$. Using a random integer $r=67$, perform the following tasks:
 1. Determine the modulus n used in the Paillier cryptosystem.
 2. Compute Carmichael's lambda function (λ) based on p and q . (5)
 3. Demonstrate the encryption of the plaintext message m using the Paillier cryptosystem with the calculated parameters, and compute the resulting ciphertext c and verify message received by sender.
- 3) Consider the ElGamal cryptosystem with the following parameters: Prime modulus $p=29$, Primitive root $g=2$ and Private key $a=7$
 - A)
 1. Encrypt the message $m=15$ using the ElGamal cryptosystem with the recipient's public key $A=12$. (5)
 2. Decrypt the ciphertext $(c_1, c_2)=(3, 23)$ using the private key $a=7$
 3. Discuss the security implications of using the ElGamal cryptosystem with small prime modulus $p=29$.
 - B) Compare and contrast the security implications of using SHA-1 versus SHA-256 in modern cryptographic applications. (5)
- 4) Suppose Alice and Bob uses trivial RSA digital signature in which they choose $p=823$ and $q=953$ and Alice sends a message $M=19070$ to Bob. Calculate the Signature and how bob verifies using public key $e=313$. (5)
 - A)
 - B) Consider a simplified key exchange algorithm with $p=37$ and $g=5$, where three (5)

parties (A, B, and C) choose private keys as follows: User A's private key (a) = 10, User B's private key (b) = 20, User C's private key (c) = 15. Calculate the shared key between them and also common shared key.

- 5) A small e-commerce company, "CleverGadgets," experiences a surge in login attempts to their customer accounts. The IDS detects a high number of failed logins originating from a single IP address.
- A)
1. What type of attack is most likely occurring, and why?
 2. Should the IDS automatically block the suspicious IP address? Explain your answer. (5)
 3. Describe the steps the security team should take to respond to this incident.
 4. How can they differentiate between a brute-force attack and legitimate login attempts?
- B) Malware targets systems directly, while firewalls focus on network traffic. Analyze how they work together for a comprehensive defense strategy. (5)

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